Element B - Sydney 2/1/24

Prior's attempts to solve the problem consisted of methods using electromagnets and theatrical representations.

Props rentals used a method where they have three different roses for each stage of the roses' life throughout the show. The petals on the rose are magatized and are attached to the bulb through electromagnetics at the top. The electromagnet is diactiviated by flipping a switch allowing the petals to fall. It comes along with scatter petals to put around the dying flower. They have lights on the bottom controlled by a remote and the electromagnetic petals are also controlled via remote.

Pro:

1. Illustration: The video of the petal falling looks very realistic and they are not dragged down.
2. Durability: Uses materials (wood, plastic, metals) that are able to withstand wear, pressure or damage.

Con:

1. Complexity: This method involves using electromagnets which are intricate to make. These electromagnets are controlled by a remote control that requires specific skills, time and varying equipment.
2. Time: Due to the time restriction of production in time for the play, this method would not be suitable.
3. Equipment: The equipment used such as electromagnets, wiring, lighting, and remote devices is costly and requires specific skills.

Krystal Myers connected an electromagnet to each petal on the rose and ran the wiring into the back of the "West Wing" through a light switch box. She powered the device using a low voltage landscape lighting power pack and turned on and off the electromagnet with a switch. The rose was lit up using a landscape light.

Pro:

1. Visual Effects: No string attached to the petals allows the petal to fall without possibilities of seeing a string attached.

Con:

1. Cost: The cost of materials is 120 dollars.
2. Writing: The wiring would have to run off stage which is unrecommended,
3. Safety: Requires electricity and wiring which proposes safety hazards of electrocution or electrical fires.

T.J. Thomas created an engineering rose by first creating a base to cover a person in and then designed the rose stem with flexible copper tube wrapped in green floral tape. Multiple thin pieces of wire ran through the copper tube and through the base to the hidden person to be pulled when needed throughout the performance. The pieces of wire ran through the copper tube to the top of the flower where they are taped to individual petals creating an even bundle at the top. When the wire was pulled, the tape would disconnect allowing the petal to fall. This design requires a stage crew or extra person to be hidden under the table for the entirety of the time that the rose is on the stage

Pro:

1. Electrical: No electrical wiring or electrical powering necessary
2. Safety: No electricity or electrical wiring allows for no risk of electrocution or electrical fires from this deceit.

Con:

1. Usage: Not compilable with actors or stage crew

Steve Murch uses an arduino board, bluetooth and iOS to build a replica of the beauty and beast rose that has functioning petals that fall and has lights lighting up the rose. He uses tiny rod magnetics mounted in the bud that are pulled by fishing line. The bulb is wooden with 4 holes and magnetic rods running up to them. The petals have small washers that connect them to the bulb when the battery is on. There is a servo motor for each magnet and when it is activated it lowers a spring touching the magnetic rod then returns to normal. This releases the petal to drop. Then, written in Swft 4 (Xcode), he uses a Macbook and an IDE XCode to publish an app to your device and uses a source code from Github. Then use Github to get the code and write it in the Arduino Sketch Editor and deploy it to the Arduino via USB cable. Use a breadboard to connect the three clusters of batteries, servo motors, resistor, arduino Neoprene Strip and RebBear BLE Shield.

Pro:

1. Application: The rose doesn't require string or wires to be run off stage and is has a simple re-set

Con:

1. Batteries: Requires 17 AA batteries and approach requires the prop to be on the entirety of the show
2. Construction: The construction of the prop requires IoT skills, specific and costly equipment, and an abundance amount of time

Work Cited

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